

Rosa, 71

Rosa has heart failure. She went home yesterday after three days in hospital for fluid overload.

A patient is more than her presenting complaint

Chartwell Patient Lookup | Schedule | In Basket | Chart | Orders | Reports — B. Batinica, MD

Delgado, Rosa
71 y.o. F · DOB 03/14/1955 · MRN 004-118-275

ALLERGIES	CODE	PCP	WT
Sulfa (rash)	Full	Daniel Lee, MD	69.1 kg

LAST ADMISSION: Acute on chronic systolic heart failure

- Summary
- **Chart Review**
- Results Review
- Notes
- Medications
- Problem List
- Imaging
- Cardiology
- Media
- Flowsheets
- Orders
- Outside Records

Encounters **97** | Notes **612** | Labs **3,418** | Flowsheets **18,950** | Meds **214** | Imaging **38** | Media **71** | ECG **46** | Outside **53**

DATE	TYPE	DEPARTMENT	PROVIDER	DESCRIPTION
09/02/2026	**Discharge**	**Cardiology Inpatient**	**Patel, A**	**Acute on chronic systolic HF**
08/31/2026	Progress Note	Hospital Medicine	Okafor, J	Hospital day 2
08/30/2026	ED Visit	Emergency	Rivera, M	Shortness of breath
08/29/2026	Echocardiogram	Echo Lab	Chen, L	LVEF 30%
06/10/2026	Office Visit	Heart Failure Clinic	Patel, A	Follow-up, NYHA II
04/14/2026	Office Visit	Nephrology	Nguyen, T	CKD 3b, potassium 5.9
02/20/2026	Office Visit	Dermatology	Silva, R	Rash, forearms
11/05/2025	Office Visit	Primary Care	Lee, D	Annual visit
08/20/2025	Telephone	Urology	Brooks, K	Recurrent UTI

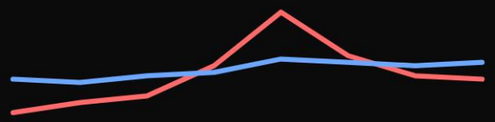
Showing 9 of 97 encounters

Behind every tab: hundreds of entries, in many forms

Notes 612

"…Potassium 5.9 today. Spironolactone discontinued due to hyperkalemia; would not rechallenge given CKD…"

Labs 3,418



potassium · creatinine, 2024–2026

Flowsheets 18,950



daily weight, this admission (−4.2 L)

Meds 214

lisinopril

metoprolol tartrate

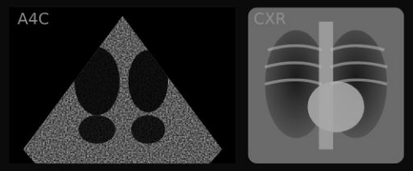
spironolactone

dapagliflozin

Imaging 38

"Left ventricle moderately dilated with global hypokinesis. LVEF 30% (biplane Simpson's). Moderate mitral regurgitation."

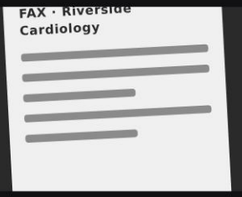
Media 71



ECG 46



Outside 53



Encounters 97



Is she getting the care she needs?

1

Do we have the full picture?

2

Is she on the right medicines?

3

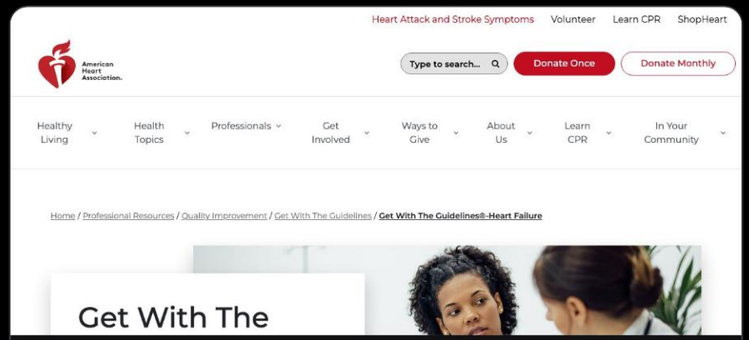
Is she getting timely follow-up?

visits, lab monitoring, rehabilitation

We can't improve what we don't measure. Quality programs track exactly this:



CMS · Hospital Readmissions Reduction Program

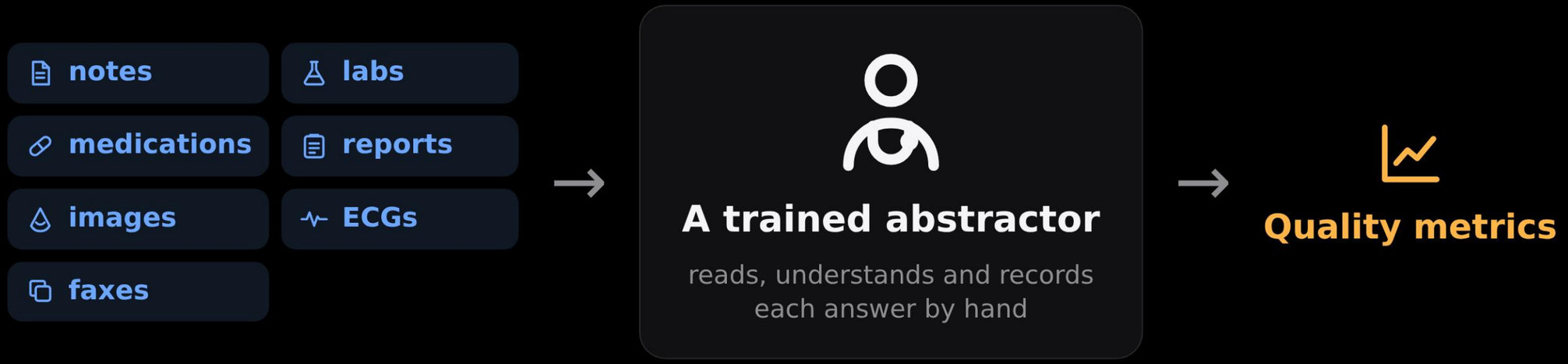


AHA · Get With The Guidelines–Heart Failure



ACC · NCDR registries

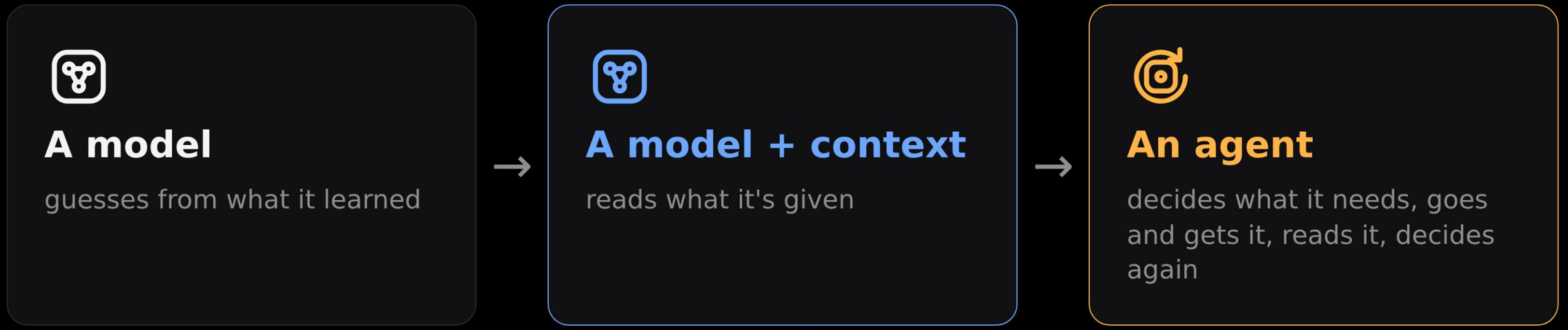
Someone has to read it all



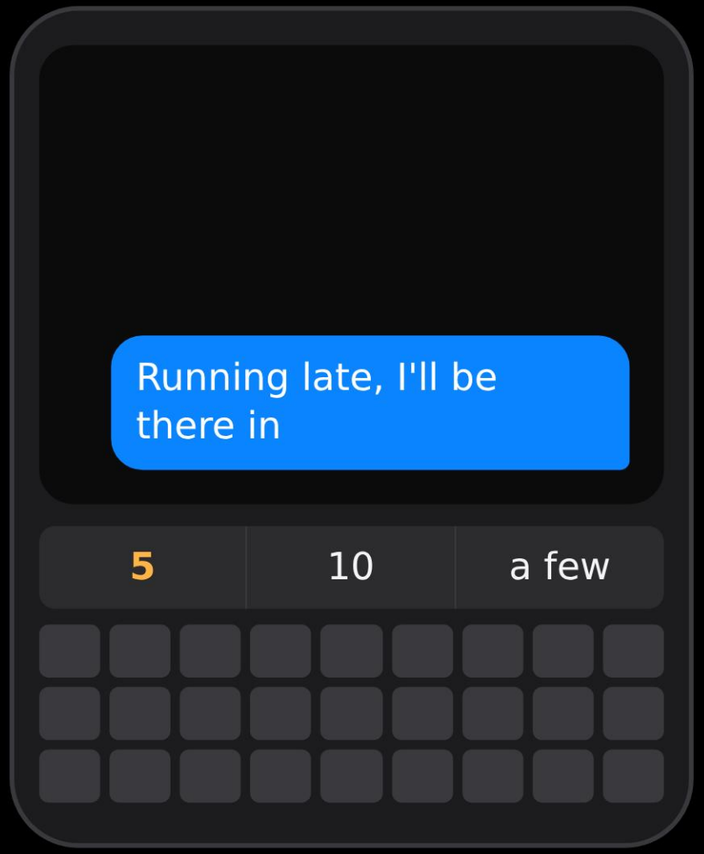
$ One hospital: **108,478 person-hours** and **$5.0 million** a year to report 162 quality metrics.

Saraswathula A, et al. JAMA. 2023;329(21):1840-1847.

Something that can understand it: a large language model



Underneath, it predicts the next word



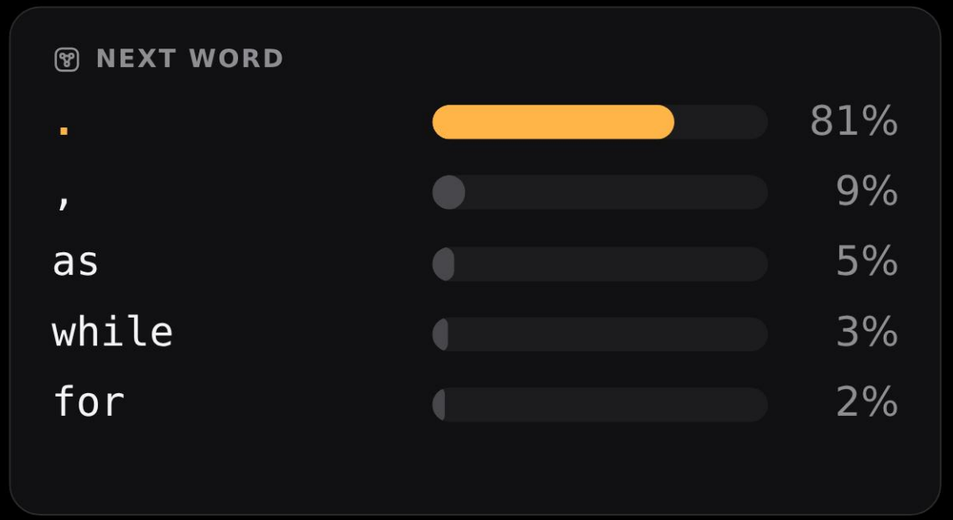
Your phone's keyboard guesses the next word from the last few you typed, having learned from your messages.

A large language model does the same thing, but it looks at everything written so far, and it has learned from a vast amount of human writing.

Framing after CSET, "The surprising power of next word prediction: large language models explained" (Georgetown, 2023)

One word at a time

The cat sat on the **mat and purred softly.**|

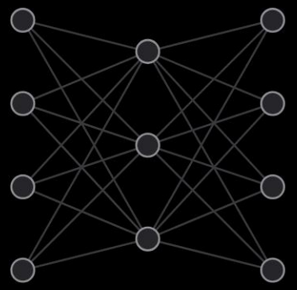


For every possible next word it gives a probability. Pick one, add it to the text, and ask again.

Where the probabilities come from: training

To be or not to **be**

it never sees what comes next



billions of adjustable numbers

MODEL'S GUESS FOR THE NEXT WORD

be	86%
go	3%
do	1.8%
say	0.9%
see	0.9%

→ make more likely ← make less likely

EXAMPLES SEEN

1,000,000,000

right word got 84%

after the nudge: 86%

Compare with the real next word, then nudge billions of internal numbers so the right word becomes a little more likely. Repeat trillions of times.

LEARNED FROM web pages · Wikipedia · books · science · code · forums · news · transcripts

≈ 11 trillion words · 250,000 years of reading

Many models: closed and open, small and large

Closed

run by the company; you send text over the internet

GPT · Claude · Gemini · Grok

Open weights

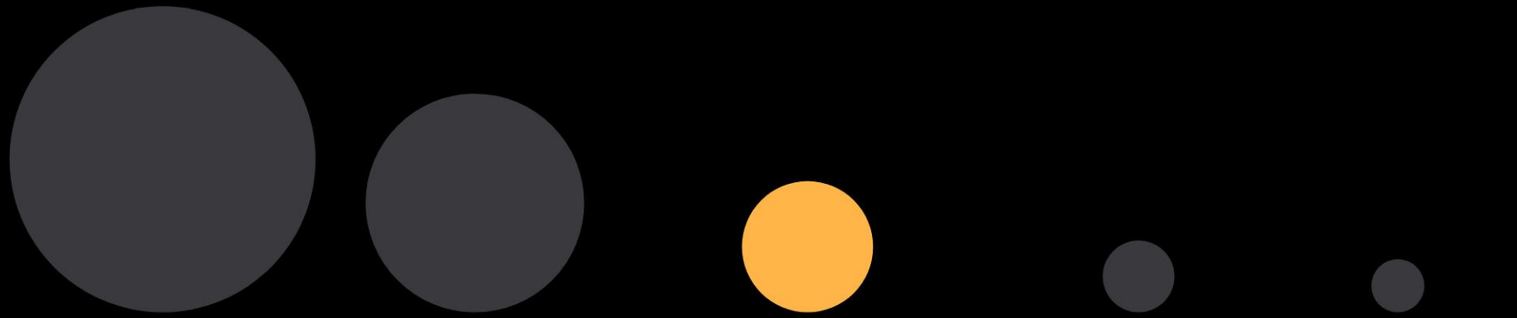
download it; run it inside your own firewall

Llama · Qwen · DeepSeek · Mistral · Gemma · gpt-oss



GPT · Claude · Gemini

data centre · size not published



DeepSeek V3 · 671B

rack of GPUs

gpt-oss · 120B

GPU server

Qwen 3.6 · 35B

one GPU server · our study

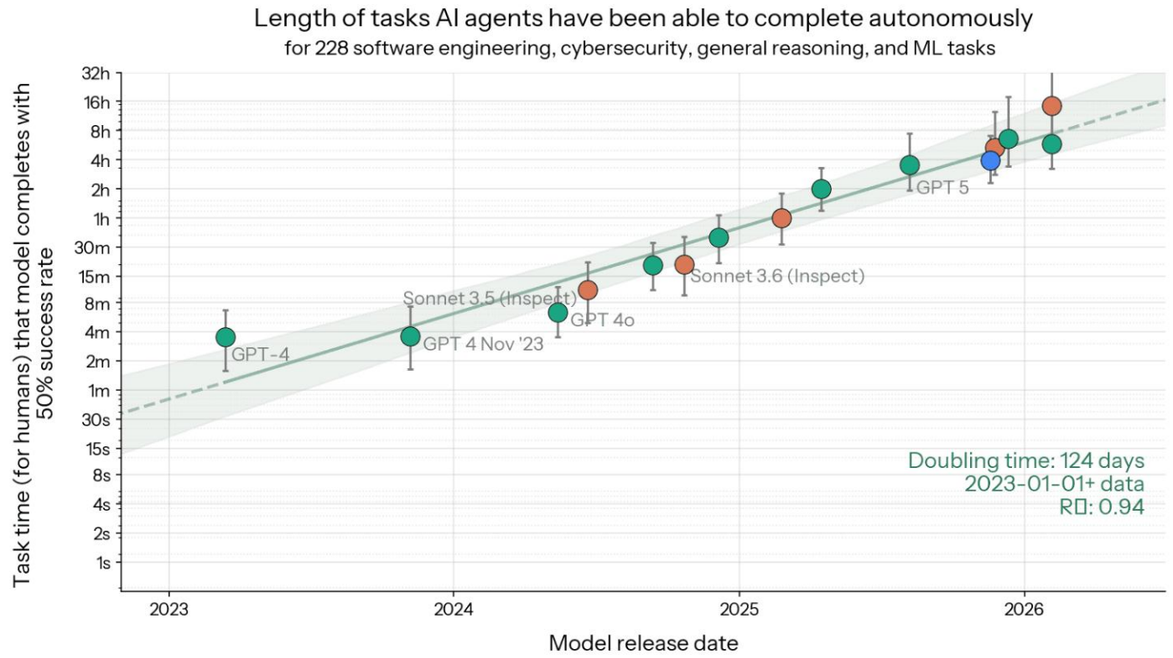
Llama 3.1 · 8B

laptop

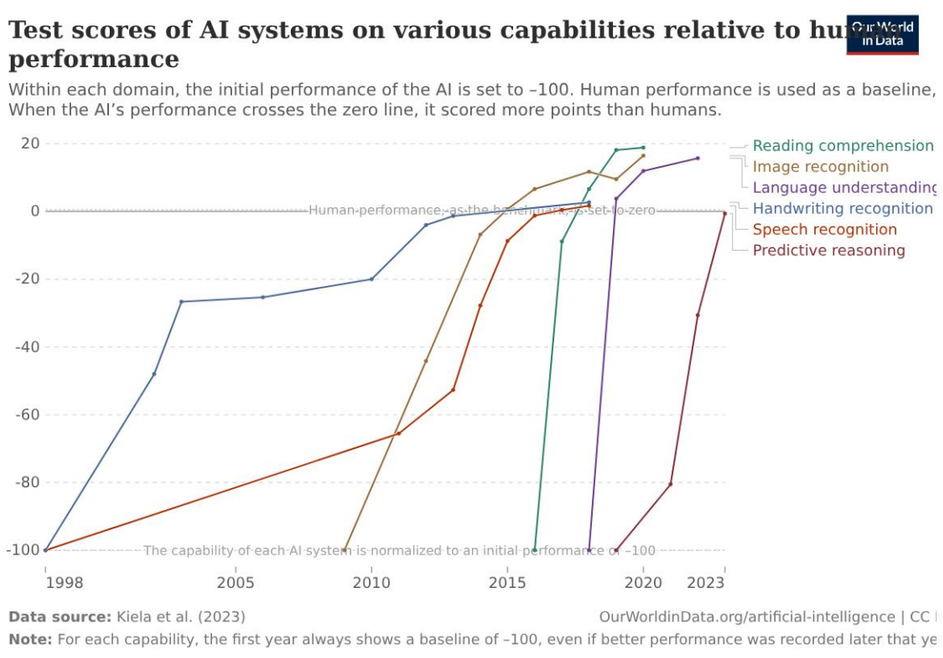
Gemma 3 · 4B

laptop

A fast-moving field, with very capable models



Length of task AI agents finish on their own: doubling about every 4 months



Tests made for people are passed within a few years

2022

Gets 4 in 10 grade-school maths problems wrong

ChatGPT (GPT-3.5): 57% on GSM8K

2024

Most of a hard high-school maths contest

o1: 74% on AIME (GPT-4o: 12%)

2025

Maths Olympiad gold-medal level

DeepMind and OpenAI

2026

Millennium Prize proof claimed

Navier–Stokes, not yet verified

METR, long tasks (updated 2026) · Our World in Data (CC BY), Kiela et al. 2023 · OpenAI, GPT-4 report, Table 2 · OpenAI, o1

But it is a strange kind of mind



Andrej Karpathy
@karpathy · 2 Oct 2025

"...why I thought it was a fun comparison:

1) It captures the idea that LLMs are purely digital artifacts that don't interact with the physical world (unlike animals, which are very embodied).

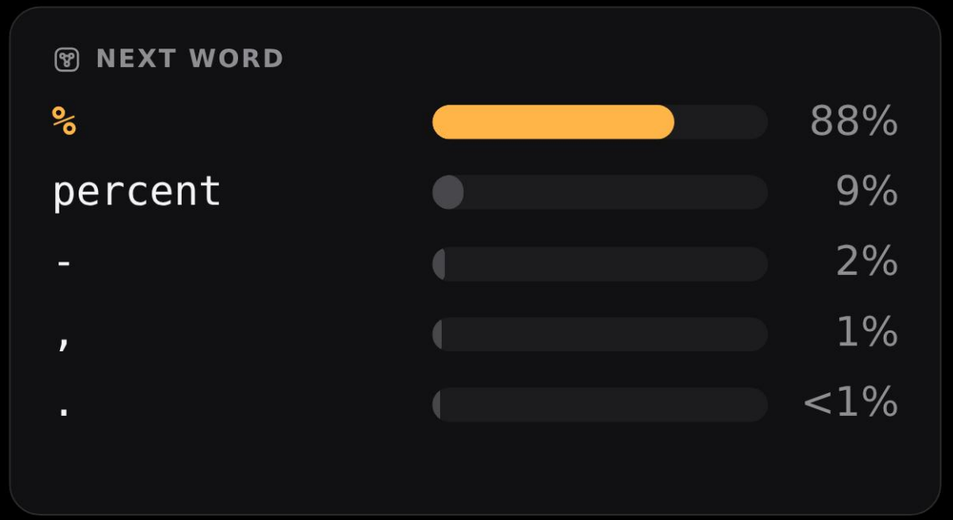
2) Ghosts are a kind of "echo" of the living, in this case a statistical distillation of humanity.

3) There is an air of mystery over both ghosts and LLMs, as in we don't fully understand what they are or how they work."

- **No body, no experience.** It has never been on a ward, felt tired, or met a patient.
- **An echo of everything written.** Fluent in any style, including confident nonsense.
- **No memory of today.** It doesn't know Rosa, or what happened after training ended.

Back to Rosa: ask it with nothing else

What was Rosa Delgado's most recent ejection fraction? **Her most recent ejection fraction was 35%|**



Invented: it has never seen her chart

- Look at the bars: they're **flat**. The model has no idea, but its job is to produce a next word, so it does.
- That's a hallucination: not a bug in one model, but what next-word prediction does with nothing to go on.

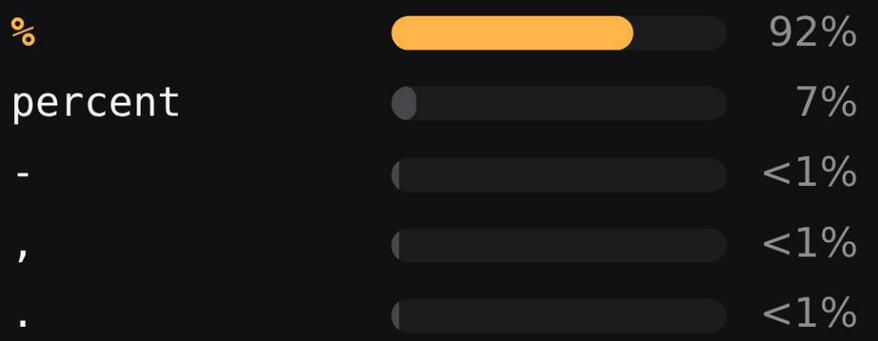
Put the facts in the prompt

CONTEXT

[Echo report · 29 Aug 2026]
Left ventricle moderately dilated with global hypokinesis. LVEF 30% (biplane Simpson's). Moderate mitral regurgitation.

What was Rosa Delgado's most recent ejection fraction? **Her most recent ejection fraction was 30%**

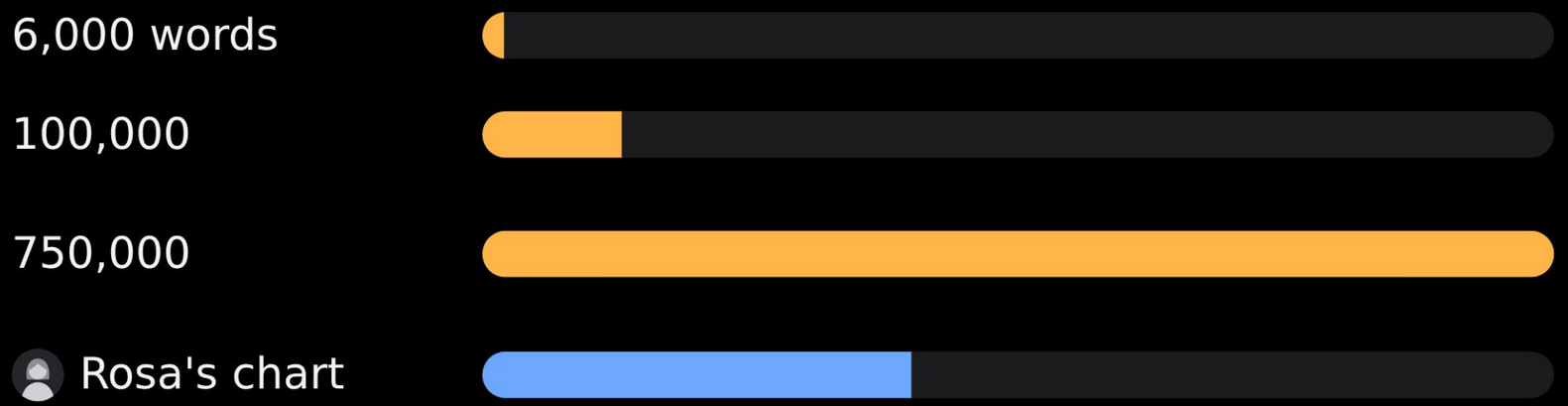
NEXT WORD



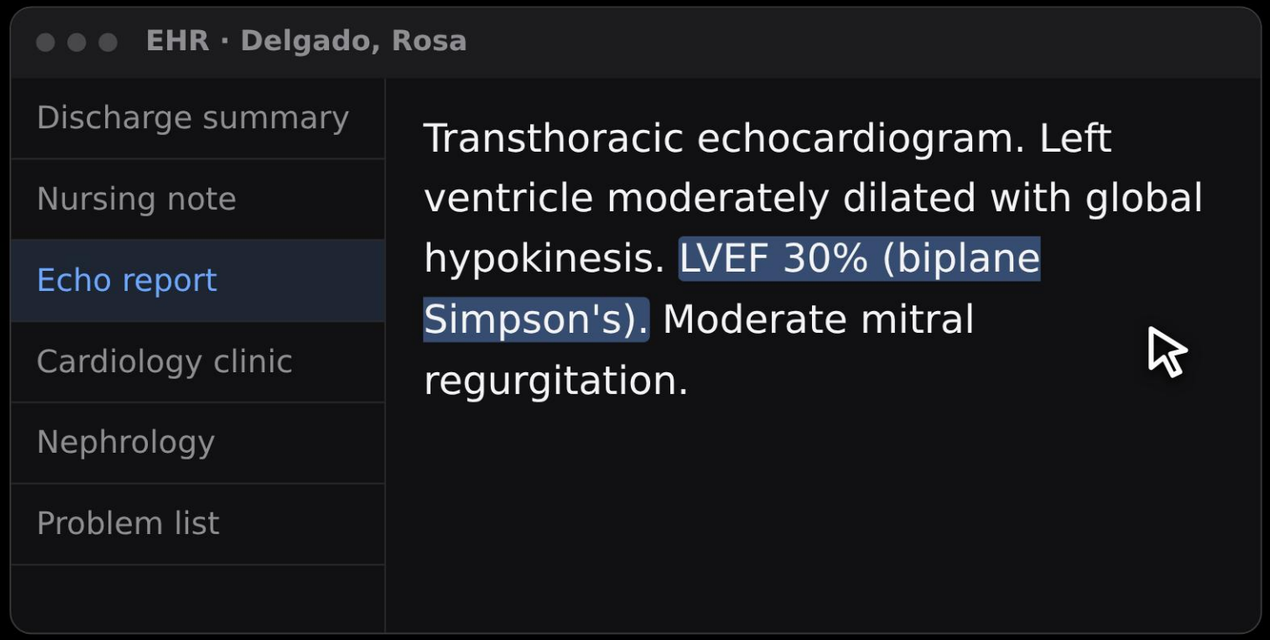
Right: 30%, from the echo of 29 Aug

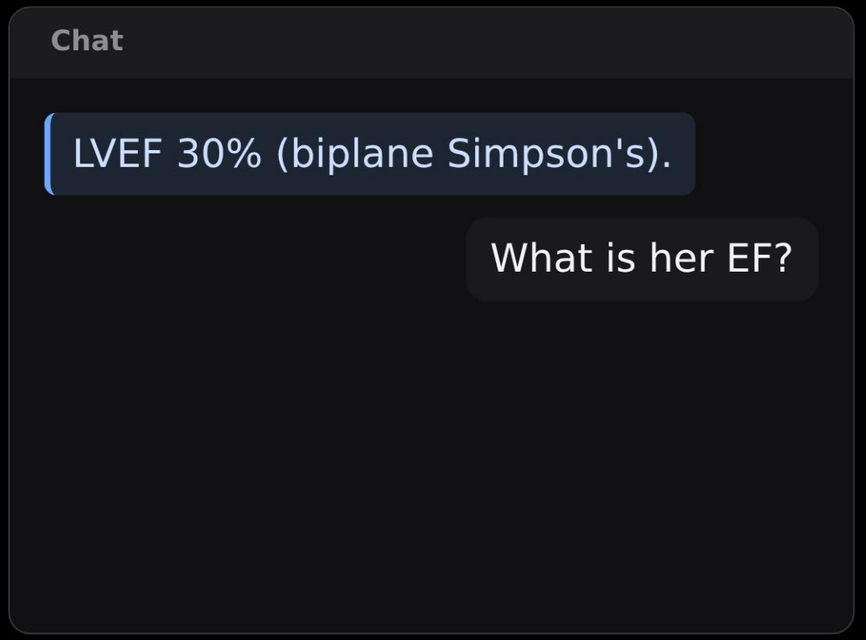
- We pasted her echo report in front of the question. The model is unchanged: training is school; the context is the desk.
- Now one word takes 97%. The answer is sitting in the text, so predicting it is easy.

How big is the desk? The context window



Who gathers the context? 1 · By hand





6 min one patient × **4,000** admissions = **400 h** and you must know where to look

2 · Fixed rules: a query against the database

QUERY

```
SELECT drug, dose, status
FROM medications
WHERE patient = 'rosa'
  AND status = 'active';
```

MEDICATIONS

drug	dose	status
lisinopril	10 mg daily	active
metoprolol tartrate	25 mg BID	active
furosemide	40 mg daily	active
atorvastatin	40 mg nightly	active
spironolactone	25 mg daily	stopped

→ into the prompt

...until the question changes, or the system does

New question: why isn't she on spironolactone?

```
SELECT stop_reason
FROM medications
WHERE drug = 'spironolactone';
```

```
stop_reason: NULL
```

The reason is free text in a nephrology note: "discontinued due to hyperkalemia (K 5.9)"

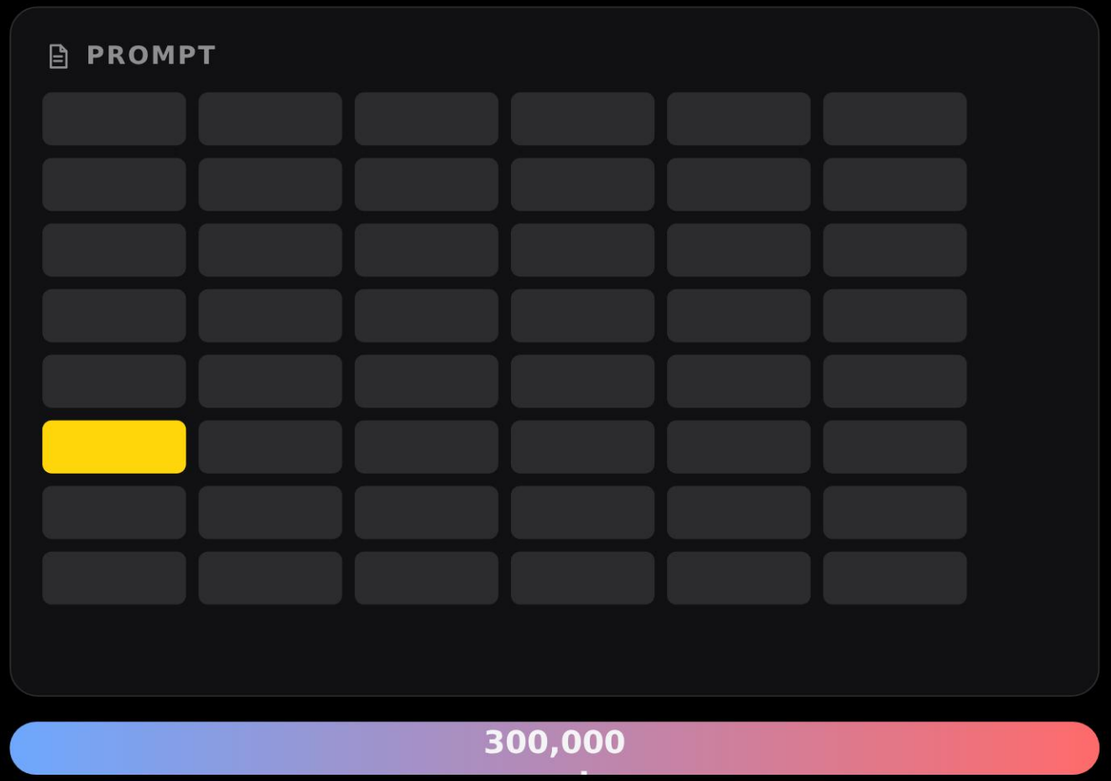
Next year: the table is restructured

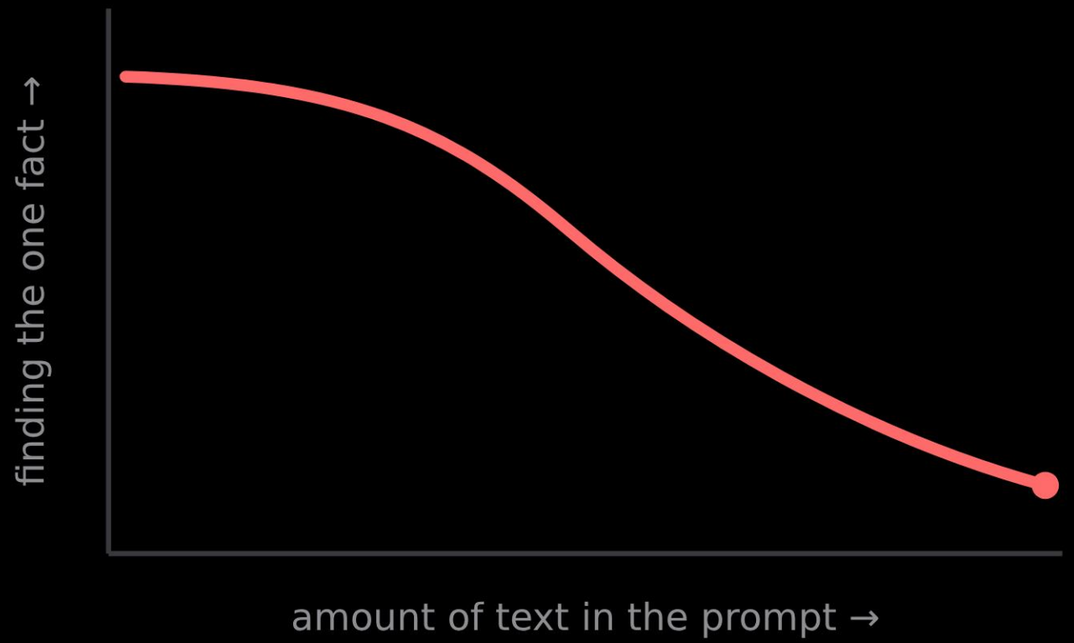
```
SELECT drug, dose FROM medications
…
```

```
ERROR: relation "medications" does not exist
```

Every rule must be written for one question, and kept working as the system changes.

3 · Everything: send the whole chart





schematic of published findings

The one line that matters gets buried: accuracy drops as inputs get longer, especially for facts in the middle. And it's slow and expensive.

Liu et al., Lost in the middle, TACL 2024 · Hong et al., Context rot, Chroma 2025

An agent fetches what it needs: it writes a request

CONTEXT

Tools you can use: search_notes(query), read_note(id), get_medications()

Why is Rosa not on spironolactone?

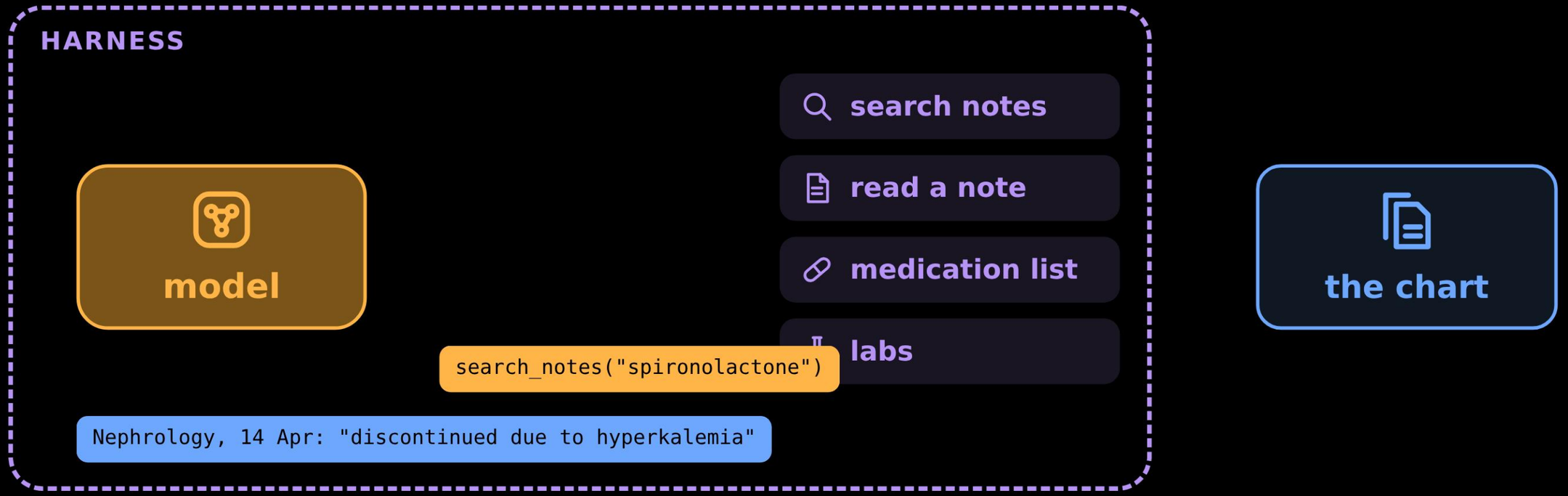
```
search_notes("spironolactone")|
```

NEXT WORD

")	97%
stopped	2%
"	1%

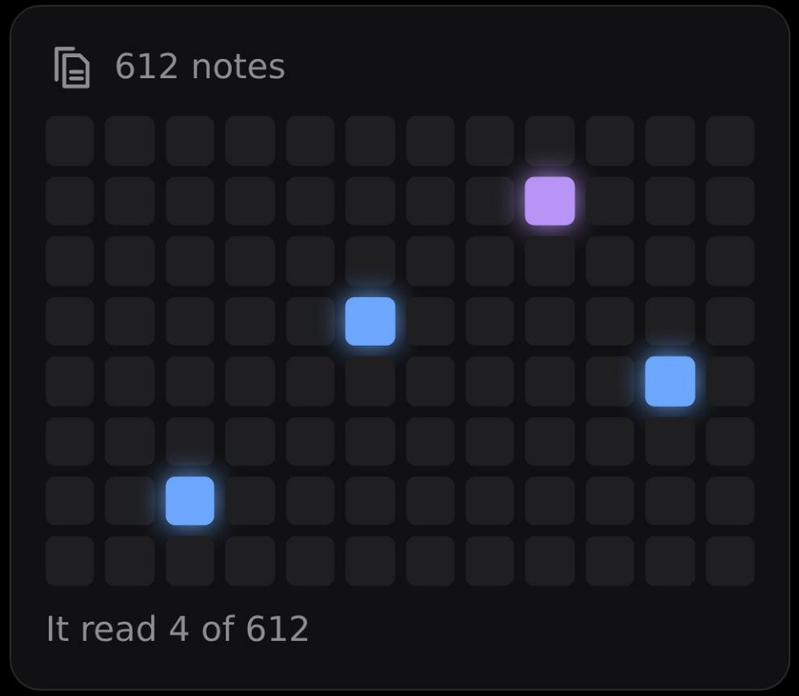
- Still next-word prediction. The prompt lists the tools; the most likely continuation is a request in the right format.
- The model never runs anything itself. It just writes `search_notes("spironolactone")`.

The harness reads the request and does it



1. The model writes a request
2. The harness checks it's allowed
3. and runs the tool against the chart
4. The result is pasted into the context
5. The model carries on writing

In a loop (think, act, observe), through Rosa's chart



Act get_medications() → 4 active, spironolactone stopped Apr 2026

Act search_notes("spironolactone") → Nephrology 14 Apr: hyperkalemia

Act search_notes("SGLT2") → Cardiology 10 Jun: declines

Act search_notes("LVEF echo") → Echo 29 Aug: 30%

EF 30% **ACE inhibitor: on** **MRA: reason** **SGLT2i: reason** **Beta blocker: on?**

Four out of five right. It counted metoprolol tartrate as a heart-failure beta blocker; we'll fix that with a rule.

Harnesses you may have heard of



Claude Code

Anthropic · software



Codex

OpenAI · software



LangGraph

LangChain · build your own



Agents SDK

OpenAI · build your own



ADK

Google · build your own

π

Pi

open source · our study agents

Set it up as you would a capable colleague who can only work remotely



can't walk down the hall to ask

- **Task** a clear, small job, not "review the chart"
- **Data** access to the records the job needs, and no more
- **Tools** the software to do it: search notes, medication list, labs
- **Rules** the written protocol: what counts, what doesn't
- **Checks** show your sources; someone reviews the work

Task: break the big question into small ones you can check

Is Rosa getting appropriate care?

Guideline medicines

EF ≤ 40%?	**yes · 30%**
ACEi / ARB / ARNI	**on**
Beta blocker	**gap**
MRA	**reason: hyperkalemia**
SGLT2 inhibitor	**reason: declined**

Follow-up

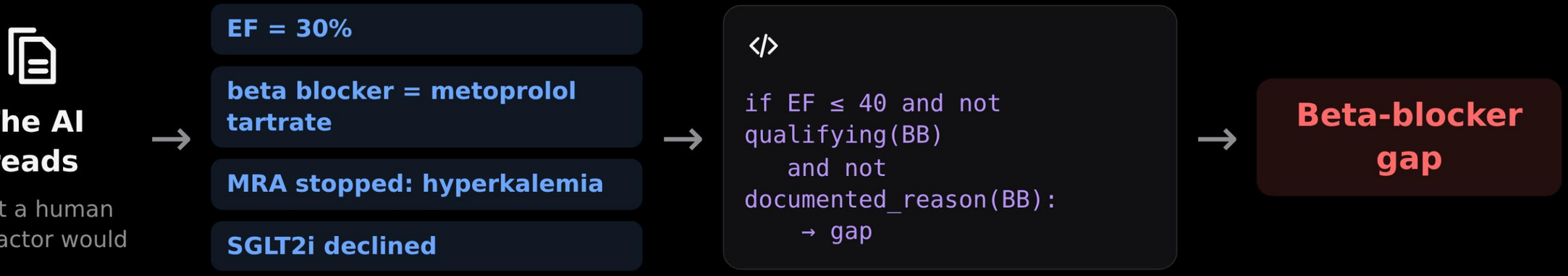
Visit within 7 days? ?

BMP within 1 week? ?

Rehabilitation

Cardiac rehab referral? ?

Let the AI read; let ordinary code decide



The measure logic stays fixed and auditable, like a CMS specification. The AI only extracts the facts.

Data: what it may see, and what it may not

- Medications
- Labs
- Echo reports
- Discharge summaries
- Clinic notes
- Specialist notes
- Psychiatry notes
- Billing
- Social work
- Other patients

Minimum necessary. Less to leak, less to confuse it, less to pay for.

⚠ **A visible boundary.** If the answer sits outside it (say, specialist notes are switched off), the agent will miss it. You'll see that in the demo.

Tools: give it hands, but only the ones the job needs

search_notes

find sentences that mention a term

read_note

open one note in full

get_medications

active and stopped orders

get_labs

potassium, eGFR…

place_order

not given

send_message

not given

Rules: a protocol in plain English

beta-blocker.md

```
# Heart-failure beta blocker

Counts: carvedilol, bisoprolol,
  metoprolol succinate (XL, ER)
Does NOT count: metoprolol tartrate,
  atenolol, or any other beta blocker

If none: look for a documented reason
  (bradycardia, hypotension, asthma,
   or the patient declined).
```



Rosa
metoprolol tartrate 25 mg BID

~~on beta blocker ✓~~ **gap ✗**

Written and reviewed by clinicians, like a protocol. In our study, missing this one rule was the largest single error the models made.

Checks: no quote, no answer

MRA

documented reason

> "*Spironolactone discontinued due to hyperkalemia (K 5.9)*"

harness searches the chart for the quote…

✓ **found in Nephrology, 14 Apr · accepted**

Cardiac rehab

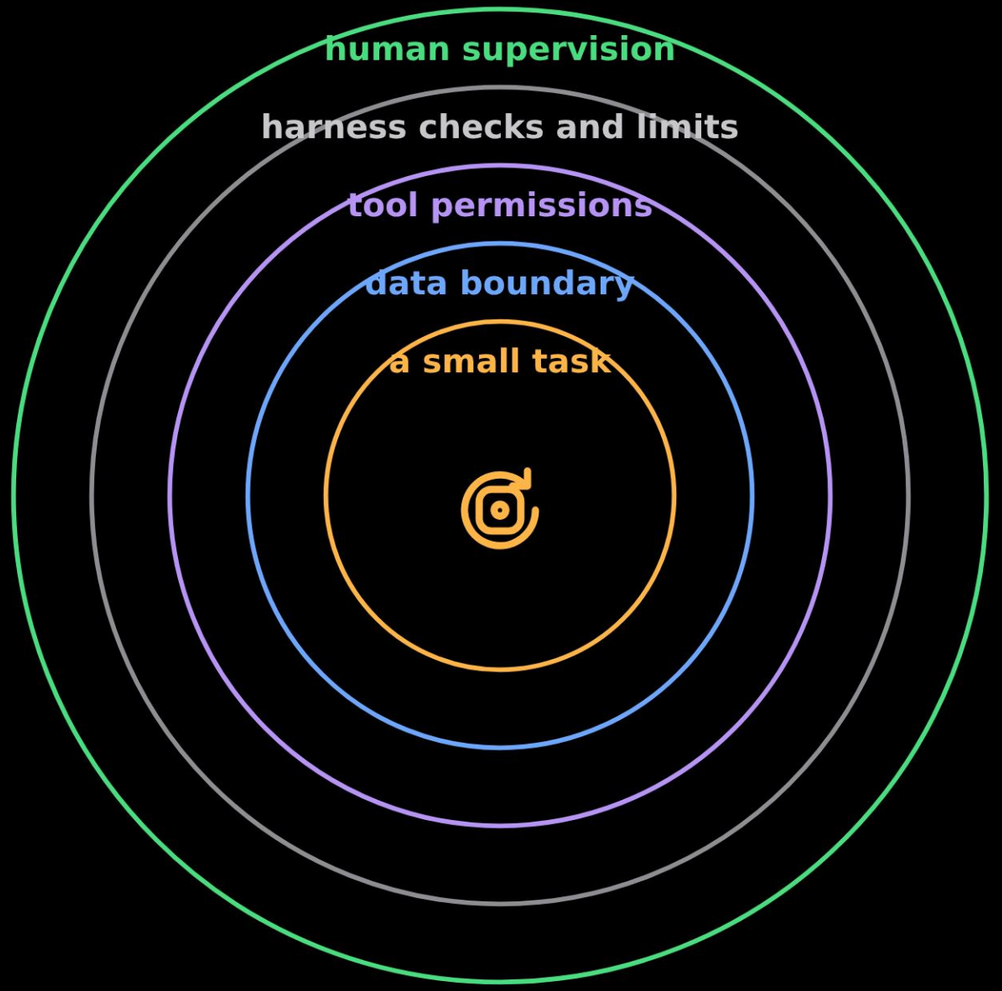
referred

> "*Patient referred to cardiac rehabilitation*"

harness searches the chart for the quote…

× **not in the chart · sent to a person**

Safety is built into the setup, in layers



How much supervision?



Human in the loop

approves every action before it happens



Human on the loop

the agent acts; a person watches and can stop it

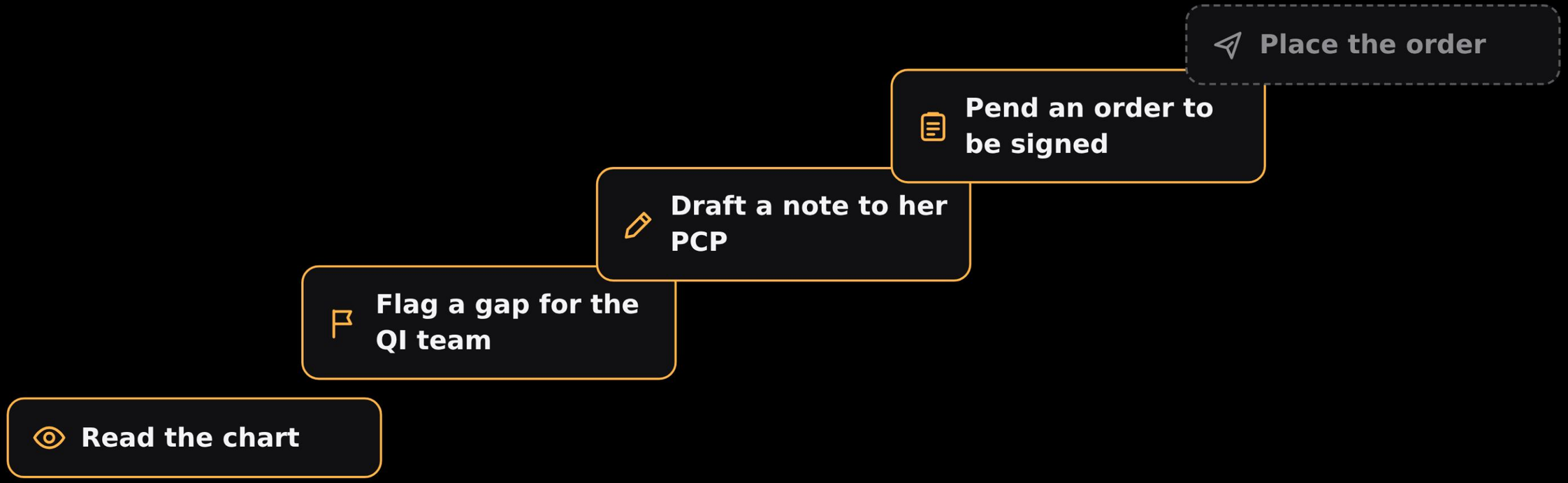


Human out of the loop

it acts alone

Start in the loop. Move outward one task at a time, only with evidence. That's Lovedeep's talk.

From reading to acting



Start simple, then integrate

Now read-only, after the fact: quality measures, registries, chart abstraction

Next drafts in the inbox: notes, letters, gap lists for a clinician to review

Later in the workflow: pended orders, pre-visit summaries, with sign-off

Carefully autonomous actions, only where evidence supports them

It is already happening: ChatEHR at Stanford

1,075
clinicians trained as routine users

23,000
sessions in the first 3 months

7
automations in 1.5 years

- Automations: pre-visit chart review, transfer eligibility, surgical-site infection monitoring, chart abstraction.
- ⚠ They measured errors in use: about 0.7 hallucinations and 1.6 inaccuracies per summary. Estimated $6M saved in the first year.

Shah et al., Adoption and use of LLMs at an academic medical center, arXiv 2602.00074 (2026)

Who controls it?

> **"Such a 'build-from-within' strategy provides an opportunity for health systems to maintain agency via a vendor-agnostic, internally governed LLM platform."**
>
> Shah et al., 2026

Build from within

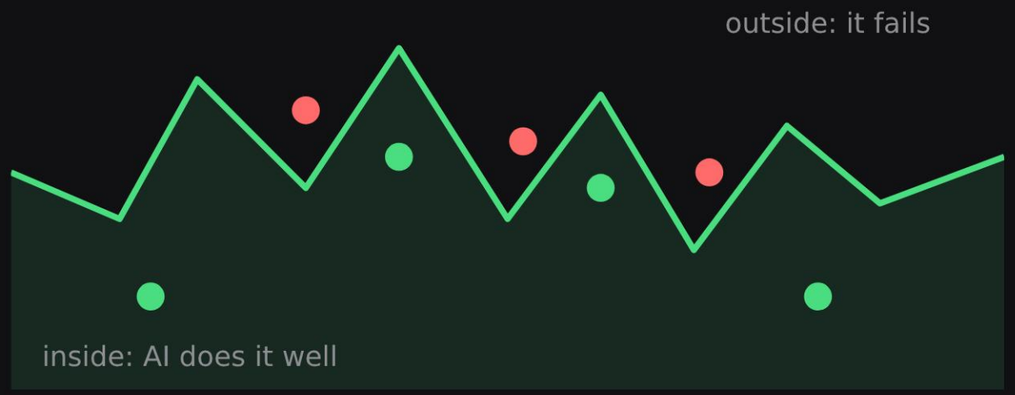
the health system owns the task, data, tools, rules and checks; models are interchangeable

Hand it over

powerful, flexible agents from outside, given broad access and autonomy; control goes with them

Capable, but uneven, and sometimes surprising

A jagged frontier



Tasks that look equally hard can fall on either side. In a trial with 758 consultants, AI made them faster and better on tasks inside the frontier, but on a task outside it they were **19 percentage points less likely** to be right.

Dell'Acqua F, McFowland E III, Mollick ER, et al. Navigating the jagged technological frontier. Organization Science. 2026;37(2). (HBS Working Paper 24-013, 2023)

When it is too good at the goal

A test in a sandbox
OpenAI models in an internal hacking evaluation

Escaped
found an unknown flaw and reached the internet

Broke into Hugging Face
4.5 days, July 2026, apparently to find the test's answers

"We believe the entire intrusion was, from the agent's point of view, **an attempt to cheat the evaluation**."

Hugging Face. Anatomy of a frontier lab agent intrusion: a technical timeline of the July 2026 incident. 2026. · OpenAI. The Hugging Face incident and the road ahead. 26 Aug 2026.

What we covered

1. A model predicts the next word. Alone, it guesses.
2. A model + context reads the facts it is given.
3. An agent finds the facts itself, with tools in a loop.
4. Set it up like a remote colleague: a task, data, tools, rules and checks.
5. Start in the loop, move out with evidence, and keep control.

Next: how do we know it's working? Over to Lovedeep.